More Value, Not More Information

Wind, Solar and the Returns to Electricity Storage Arbitrage in Europe

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Abstract

Grid-scale storage in Europe is financed on the expectation that renewables create arbitrage value. We ask whether they do, and whether that value can be captured without forecasting. Using hourly day-ahead prices for 18 European bidding zones from 2019 to 2026, we operate a price-taking battery four ways that differ only in information: perfect foresight, a typical-day schedule learned from the previous four weeks, persistence, and a forecaster built on TSO day-ahead fundamentals. The treatments are daily forecast wind and solar penetration, under zone-by-year, zone-by-month and date fixed effects. Solar creates arbitrage value: 10 percentage points more forecast solar raises a day's perfect-foresight value by 28%. Wind creates none. Solar also moves the day's price peak and trough towards the hours the calendar predicts. But neither technology changes the value that requires information. The calendar operator's euro shortfall does not rise with wind, and a forecaster's advantage over it falls on windy days. These results hold when forecast penetration is instrumented with realised weather, and for a second forecaster aimed at the day's price shape. Our pre-registered test, that calendar operators capture a larger share of solar-driven than wind-driven value, has the predicted sign but is inconclusive (wild-bootstrap $p = 0.12$). A registered result that wind's cost grows with a system's wind share rests on a single zone. The renewable mix changes how much day-ahead storage value there is, not who can capture it. Hypotheses were pre-registered; analyses added after the first results, including in response to review, are reported as exploratory.

1 Introduction

Grid-scale batteries are being built across Europe on the expectation that the price volatility created by wind and solar will pay for them. That expectation has two parts that are usually run together. The first is that renewables create arbitrage value: they depress prices when they produce and leave scarcity when they do not, widening the intraday spreads a battery trades on. The second, rarely stated, is that the value can be captured by whoever owns the battery. Arbitrage value exists only in hindsight. An operator commits its schedule in the day-ahead auction before prices are known, and what it earns depends on how much of the day's price shape it can anticipate.

This paper asks both questions of the same data, and asks them separately for wind and solar. We had a hypothesis about the second. Solar output follows the sun, so the value it creates should recur at the same clock hours and be found by a schedule learned from recent weeks. Wind output follows weather systems that arrive at no particular hour, so the value it creates should sit wherever the weather puts it and reward operators who forecast the day. If so, a system decarbonising

through wind would reward a different kind of storage operator from one decarbonising through solar.

We test this on hourly day-ahead prices for European bidding zones from 2019 to 2026. The panel spans Danish and German wind, Iberian and Greek solar, French nuclear, Nordic hydro and Polish coal. We model a price-taking 1 MW, 2-hour battery operated four ways, which differ only in the information used to set the schedule:

1. **Perfect foresight**, which defines the ceiling.
2. **Typical day**, which optimises against the mean price profile of the previous four weeks, and so knows the calendar and nothing about the day.
3. **Persistence**, which optimises against yesterday's prices.
4. **Forecaster**, which adds a gradient-boosted correction built from the TSOs' day-ahead wind, solar and load forecasts.

Our treatments are the TSOs' day-ahead forecasts of wind and solar output as a share of forecast load. Zone-by-year, zone-by-month-of-year and date fixed effects remove capacity build-out, each zone's seasonal cycle and every shock common to Europe on a day. What remains is mostly day-to-day weather. It is not weather alone. Section 6 sets out what else it contains, and instruments the forecasts with realised weather. The design responds to the central difficulty of the question: renewable penetration trends upward everywhere, alongside everything else that changed over the decade, and a specification that lets that trend through will find an association between any two rising series. The hypotheses, specification and robustness set were registered before any backtest ran. A referee's report on the first draft found an error in our bootstrap and several overstatements; the corrected results are reported here, and every analysis added after the first results is labelled exploratory.

The main findings are these.

- **Solar creates arbitrage value; wind does not** (registered outcome, no registered direction). A day with 10 percentage points more forecast solar offers 28% more perfect-foresight value, about €48 per MW per day, and a €22/MWh wider spread. A windier day offers no more value and a slightly narrower spread. The solar result survives every identification check we or the referee devised, including instrumenting the forecasts with realised weather. It shrinks by about 40% when zones are compared only with their own neighbours.
- **Solar makes the day's price shape more typical, and wind makes it less typical** (registered, H2). Solar also moves the day's peak and trough towards the hours the calendar predicts. The wind effect is smaller than it first appears: about 60% of it is explained by the next day's wind, and its placebo test fails.
- **The value that requires information does not change with the renewable mix** (exploratory). The calendar operator leaves about €31 per MW per day on the table. Wind does not increase that shortfall, and solar, if anything, reduces it. Solar raises the calendar operator's revenue one-for-one with the ceiling. The forecaster's advantage over the calendar does not rise on windy days; it falls. A second forecaster aimed at the day's price shape finds no premium to

wind either.

- **Our pre-registered test is inconclusive** (H1). The calendar operator's *share* of the ceiling rises more with solar than with wind, by 4.4 points per 10 pp, with a wild-bootstrap p-value of 0.12. Given the third finding, the share difference is mostly a restatement of the first: solar adds value the calendar can find, and wind adds none.
- **A registered heterogeneity result rests on one zone** (H4). Wind lowers calendar capture more in wind-heavy systems, but only because of Denmark West. Without it the interaction is imprecise and of either sign.

So the renewable mix changes how much storage value there is. We find no evidence that it changes who can capture it. For the day-ahead market, the premise that wind-driven systems reward forecasting sophistication is not supported by these data, and the premise that solar creates capturable value is.

The paper makes three contributions:

1. **It separates the size of storage value from its accessibility.** Existing work computes the perfect-foresight value of arbitrage across European zones (Mercier, Olivier, and De Jaeger 2023), or measures forecast-based capture within one or two markets (Hornek et al. 2025; Falezza 2026). Combining an operator ladder with cross-market variation lets us ask how each part responds to the generation mix.
2. **It extends to storage the evidence that wind and solar shape prices asymmetrically** (Hirth 2013; López Prol, Steininger, and Zilberman 2020; Bushnell and Novan 2021), with identification from forecast penetration under date fixed effects across 18 zones.
3. **It reports a pre-registered test honestly, including the parts that failed.** The registration, deviation log, referee report and corrections are part of the public record, and the paper is compiled from the code that produces every number in it.

Section 2 reviews related work. Section 3 describes the data and the operators, and Section 4 the empirical strategy. Section 5 presents results, Section 6 robustness and limitations, and Section 7 concludes.

2 Related literature

This paper connects three literatures that have developed largely separately: the valuation of storage arbitrage, the effect of variable renewables on wholesale price formation, and electricity price forecasting.

Storage arbitrage valuation. The standard approach values a price-taking store by optimising its dispatch against historical prices under perfect foresight. Sioshansi et al. (2009) apply it to PJM and also estimate the welfare effects of storage at scale; Walawalkar, Apt, and Mancini (2007) value energy arbitrage and regulation services in New York; and Sioshansi (2010) examine how ownership structure changes the welfare calculus. For Europe, Zafeirakis et al. (2016) backtest arbitrage strategies across day-ahead markets. They find value concentrated in less integrated, import-dependent

and less competitive markets, which attributes the cross-market dispersion to market structure. Closest to our setting, Mercier, Olivier, and De Jaeger (2023) compute the perfect-foresight value of a price-taking battery for every EU-28 bidding zone, plus Norway, Switzerland and Turkey, varying efficiency and duration. Our ceiling V^* is their object. We take it as the denominator and ask what share of it operators with less information realise.

A smaller and more recent literature measures that share directly, within a single market. Hornek et al. (2025) find that a forecast-driven battery captures around 89% of perfect-foresight revenue in the German continuous intraday market. Falezza (2026), trading German and Swiss day-ahead, intraday and reserve markets, finds that capture depends on the rank correlation between forecast and realised prices rather than on conventional error metrics. Forecasts above a high rank-correlation threshold capture almost all of the perfect-foresight revenue. His persistence benchmark captures about a third, but it trades several markets at once, so it is not comparable with our day-ahead persistence operator. This is the closest single-market counterpart to our mechanism, and it is why we report a rank-based version of our shape-novelty measure alongside the Pearson one. What neither paper can do with one or two markets is ask how capture varies with the generation mix, which is the variation our panel supplies. Rules that dispatch on recent historical prices have long served as benchmarks in this literature. Sioshansi et al. (2009) evaluate such a "backcasting" rule in PJM and find that it recovers a large share of the perfect-foresight value. Our typical-day operator is a rule of this kind, and we claim no novelty for it.

Renewables and price formation. Variable renewables enter the merit order at near-zero marginal cost and depress prices when they produce (Sensfuß, Ragwitz, and Genoese 2008). Regression evidence for Germany puts the effect at roughly €1 per MWh for each additional GWh of wind and solar (Cludius et al. 2014). Ketterer (2014) shows that German wind lowers the price level but raises the *day-to-day* volatility of daily prices. Intraday shape is a different object, and it is what arbitrage earns from. On that margin the evidence is technology- specific. Bushnell and Novan (2021) show that solar lowers midday prices and raises shoulder-hour prices, deepening the intraday profile. Rintamäki, Siddiqui, and Salo (2017) find that wind *reduces* daily price volatility in Denmark but increases it in Germany, and that solar reduces it in Germany. Wozabal, Graf, and Hirschmann (2016) find non-monotone effects of renewables on price variance. Because renewables depress prices precisely when they produce, their own market value falls with penetration (Hirth 2013). In California, wind and solar cannibalise each other asymmetrically: wind penetration lowers solar's value factor while solar penetration raises wind's (López Prol, Steininger, and Zilberman 2020). Our first result, that solar raises daily arbitrage value while wind narrows the spread, sits squarely within this evidence. The panel's contribution on this margin is identification across 18 zones from day-to-day weather under date fixed effects. The merit-order literature is concerned mainly with the *level* of prices. Our outcomes depend on their intraday shape, and specifically on whether that shape is predictable from the calendar.

Renewables and the value of flexibility. If renewables raise volatility, they should raise the value of storage, and equilibrium studies confirm that storage investment responds to renewable-driven volatility. Butters, Dorsey, and Gowrisankaran (2025) model battery entry in California, finding it

economic without subsidy only at high renewable shares, with the first tranches of storage doing most to reduce prices. Karaduman (2023) finds a non-monotonic relationship in South Australia: storage first erodes renewable revenues by flattening prices, then raises them once renewable capacity is large enough to face curtailment. Hydro reservoirs have long performed this role. Danish exports track Nordic reservoir levels rather than wind directly (Green and Vasilakos 2012), and wind shifts the shadow value of stored water in Norwegian hydro areas (Mauritzen 2013). These studies establish that renewables change *how much* flexibility is worth. Our question is *who can capture it*: whether the value renewables create is available to any operator who knows the season, or only to one who knows the day.

Price forecasting. Electricity price forecasting is a mature field (Weron 2014). Recent work stresses rigorous, comparable benchmarking (Lago et al. 2021), and shows that European forecasts improve when neighbouring coupled markets are included (Lago et al. 2018). That last finding matters for identification here. Under market coupling, a zone's price responds to its neighbours' weather, so our wind coefficient measures the effect of wind weather in a zone's region rather than of strictly domestic output. We return to this in Section 6. We use forecasting instrumentally. Our forecaster exists to measure how much of the value lost by a calendar-based operator is recoverable with information available at gate closure, not to compete on forecast accuracy.

Identification from weather. Using weather-driven variation in wind output to identify its effects on the power system has a well-established precedent (Cullen 2013; Novan 2015). We apply the same logic to forecast penetration, with date fixed effects to remove common shocks.

Inference. Our design has a moderate number of natural clusters: 18 bidding zones in the estimation sample. They are also heterogeneous: some zones have almost no solar variation. Cluster-robust inference is unreliable in this setting (Cameron, Gelbach, and Miller 2008; MacKinnon and Webb 2017; MacKinnon, Nielsen, and Webb 2023). We report wild cluster bootstrap p-values with six-point weights (Webb 2023), and, for the key test, jackknife standard errors and the exact Rademacher enumeration. Our treatments are continuous, not staggered binary adoptions, so the negative-weighting problems of two-way fixed effects under staggered timing (de Chaisemartin and D'Haultfœuille 2020; Goodman-Bacon 2021) do not arise in their canonical form. Heterogeneity in effects across zones remains a concern, which we address with leave-one-zone-out estimates.

Data. The ENTSO-E Transparency Platform is the only harmonised public source for this panel, and it has documented quality problems: inconsistent labelling, gaps and revisions (Hirth, Mühlenpfordt, and Bulkeley 2018). We report our coverage audit and exclusions in full in Section 3.

3 Data and design

3.1 Market data

All market data come from the ENTSO-E Transparency Platform, which publishes standardised data for European bidding zones under Commission Regulation (EU) No 543/2013. We use five series at hourly resolution for 2019-01-01 to 2026-09-20:

- day-ahead clearing prices;
- actual total load;
- actual generation by production type;
- the transmission system operators' (TSOs') day-ahead load forecasts;
- the TSOs' day-ahead forecasts of wind and solar generation.

The unit of observation is the **bidding zone**, not the country, because day-ahead prices clear at zone level. Denmark's two zones sit on different synchronous areas and frequently clear at different prices.

The operators are run on 19 zones spanning the range of European generation mixes:

Group	Zones
Nordic and Baltic	DK_1, DK_2, NO_2, SE_3, SE_4, FI, EE
Central-western	DE_LU, NL, BE, FR, AT, CH, CZ, PL
Iberian	ES, PT
Southern	GR, IT_NORD

Great Britain is excluded because the platform holds no GB day-ahead prices after 2020. Ireland (SEM) is excluded because its load forecast is almost entirely absent after mid-2021. Czechia publishes no wind forecast, so its wind penetration is undefined and it drops out of every regression. **All regressions therefore use 18 zones.** Czechia remains in the descriptive statistics.

Two features of the data require care:

- **Resolution.** From October 2025 the day-ahead auction moved to a 15-minute market time unit in most zones. We average quarter-hours into hours, which is the price a battery holding a flat position over the hour receives.
- **The trading day.** Every coupled zone trades a *delivery day* defined on Central European Time, including zones whose civil time differs. Because EU daylight saving is synchronised, Portugal's trading day begins at 23:00 local time all year, and Finland's, Estonia's and Greece's at 01:00. We verified this against the platform's own price documents. Every "day" in this paper is a CET delivery day: 24 hours, or 23 or 25 on clock-change days.

3.2 Treatment variables

Our treatments are the TSOs' day-ahead forecasts. For zone z and delivery day d:

$$\text{wind}_{zd} = 100 \times \frac{\sum_{h \in d} \widehat{W}_{zh}}{\sum_{h \in d} \hat{L}_{zh}}, \qquad \text{solar}_{zd} = 100 \times \frac{\sum_{h \in d} \hat{S}_{zh}}{\sum_{h \in d} \hat{L}_{zh}},$$

where $\widehat{W}$, $\hat{S}$ and $\hat{L}$ are forecast wind output, solar output and load in MWh. A penetration is defined only when the forecasts cover every hour of the day and the load forecast is plausible. Eleven days,

all reporting errors, fall outside 50–200% of the zone's median for their month and are excluded.

The TSO forecast is a proxy for the information on which the auction clears, not a measure of it. Commercial participants use their own forecasts. The regulation also requires wind and solar forecasts only by 18:00 on D−1, six hours after the 12:00 auction, so a published forecast may contain updates the auction did not have. Two further features of the construction matter for interpretation (Section 6):

- the load denominator itself falls on sunny days;
- the treatments are compared across regions on the same day.

3.3 Arbitrage operators

We model a price-taking battery with 1 MW of power, 2 MWh of energy and 85% round-trip efficiency. Batteries of one and four hours are robustness checks. Each day it starts and ends empty, which makes days independent and comparable. For any price vector $\pi = (\pi_1, \dots, \pi_H)$ over the H hours of a day, the schedule solves the mixed-integer linear programme

$$\max_{c,s,e,z} \sum_{h=1}^{H} \pi_h (s_h - c_h) \quad \text{s.t.} \quad e_h = e_{h-1} + \sqrt{\rho}\, c_h - s_h/\sqrt{\rho},\ 0 \le e_h \le \bar{E},\ 0 \le c_h \le \bar{P} z_h,\ 0 \le s_h \le \bar{P}(1 - z_h),\ e_0 = e_H = 0,$$

with c charging and s discharging power, e the stored energy, ρ the round-trip efficiency, and binary z_h preventing simultaneous charging and discharging. The binaries are necessary rather than cosmetic. At negative prices, charging and discharging at the same time is strictly profitable in the linear relaxation, because it burns energy the operator is paid to take.

The operators differ only in the price vector they optimise against. Each schedule is then settled at the day's actual prices.

Operator	Optimises against	Information used
Perfect foresight (PF)	Actual prices of day d	The ceiling, V^*_{zd}
Typical day (TD)	Mean price by market hour over days $d-28$ to $d-1$	The recent calendar shape
Persistence (PS)	Day $d-1$'s prices, by market hour	Yesterday
Forecaster (FC)	TD profile plus a gradient-boosted correction	TD's set, plus TSO day-ahead load, wind and solar forecasts

For TD, at least 20 of the 28 days must be complete. TD and PS use only prices known at gate closure: day $d-1$ cleared on $d-2$. FC is refitted monthly on an expanding window, and its wind and solar inputs carry the 18:00 timing caveat above (details in Appendix B). Because every operator shares one optimiser, differences between them reflect information and how it is used.

For FC this is a statement about one model, not about the information itself. A forecaster trained on squared error need not dominate TD in arbitrage revenue, and it does not in four zones. So FC − TD measures how well this model uses day-specific information, not what that information is worth. Rules that dispatch on recent historical prices are a natural benchmark, and we claim no novelty for TD itself. Our interest is in how its performance varies with the renewable mix.

3.4 Outcomes

- **Capture shares**, $100 \times V^k_{zd}/V^*_{zd}$, for each operator. They are defined only where $V^* > 0$ and V^* is at or above the zone's 5th percentile of positive values. $V^* = 0$ does not mean flat prices. It means no intraday spread covered the 15% round-trip loss, which happens on 1,204 zone-days across 14 zones, 609 of them in NO_2.
- **Scale:** $\log V^*$ (registered). It drops the zero-value days, so we also report $\log(1 + V^*)$ and V^* in euros.
- **Mechanism:**
  - *Shape novelty*, $1 - \text{corr}(\pi_d, \bar{\pi}^{TD}_d)$ (registered);
  - a rank-correlation version, and the hour distances between the day's actual and typical maximum and minimum (exploratory), which separate timing from amplitude;
  - the daily spread, $\max_h \pi_h - \min_h \pi_h$.
- **Value of information** (exploratory): the value each operator leaves on the table in euros, $V^* - V^k$, and the forecaster's gain over the calendar, $V^{FC} - V^{TD}$. These avoid the capture share's near-zero denominators.

4 Empirical strategy

We estimate

$$Y_{zd} = \beta_w \, \text{wind}_{zd} + \beta_s \, \text{solar}_{zd} + \alpha_{z,y(d)} + \gamma_{z,m(d)} + \lambda_d + \varepsilon_{zd}. \tag{1}$$

The fixed effects are additive:

- **zone-by-year**, $\alpha_{z,y}$, absorbing capacity build-out, market reforms and zone-specific trends;
- **zone-by-month-of-year**, $\gamma_{z,m}$, absorbing each zone's average seasonal cycle;
- **date**, λ_d, absorbing everything common to all zones on a day, such as gas and carbon prices.

The identifying variation is a zone's day-to-day deviation from its year level and average seasonal profile, relative to the other zones on the same day. Two things follow from that description. The effects are additive, so a solar season whose amplitude grows with installed capacity is not fully absorbed; we report zone×year×month effects as a check. And date effects remove only what is common to *all* zones, so part of the variation compares different regions on the same day. We therefore also report region×date effects, which compare each zone only with its own neighbours.

We describe the variation as driven by weather, with the caveats set out in Section 6.

The registered test (H1) is $\beta_w < \beta_s$ for typical-day capture. We re-parameterise (1) with wind_{zd} and $\text{vre}_{zd} = \text{wind}_{zd} + \text{solar}_{zd}$ as regressors, so the coefficient on wind is $\beta_w - \beta_s$.

Inference. Errors are clustered by zone. With 18 clusters, conventional cluster-robust inference over-rejects (Cameron, Gelbach, and Miller 2008; MacKinnon and Webb 2017), so key p-values come from a wild cluster restricted bootstrap with Webb six-point weights (Webb 2023) and 9,999 draws. The date effects are not nested within the zone clusters, so each bootstrap draw re-projects the reweighted residuals off the fixed effects. An earlier version of our code omitted that step, which understates p-values. It was corrected after review (Appendix A). For the key test we also report jackknife (CV3) standard errors and the exact Rademacher bootstrap enumerating all 2^{18} sign patterns (MacKinnon, Nielsen, and Webb 2023).

Placebos. The realised forecast error in wind (registered) is added to (1). It is not known when day d's auction clears, so a coefficient distinguishable from zero would signal a problem with timing or measurement. We also add the next day's forecast penetration (exploratory). We first intended it as a second placebo, but weather forecasts for $d + 1$ do exist at gate closure, so it turns out to test intertemporal dependence instead (Section 6).

The hypotheses, specification and robustness set were registered before any panel backtest ran (commit `6aa7cdc`). Five deviations were logged before the first regression, two of them after the ceiling V^* had been computed. Analyses added after the first results, most at a referee's request, are labelled exploratory throughout. Appendix A gives the full record, including two corrections to the deviation log.

5 Results

Registered results are reported first in each subsection, and exploratory analyses are labelled as such. Unless stated otherwise, effects are per 10 percentage points of forecast penetration, and p-values come from the wild cluster restricted bootstrap.

5.1 How much value there is, and who captures it

Table 1 summarises the panel. A 1 MW, 2-hour battery with perfect foresight of day-ahead prices would have earned €153 per day on average across the 19 zones, or about €56,000 per MW per year from day-ahead arbitrage alone. This is a ceiling, not a forecast of revenue. The ceiling varies more than threefold across zones. It is lowest in Norway's reservoir-hydro zone NO_2 (€68), where hydro already smooths prices, and highest in Estonia (€240) and Greece (€208).

$V^* = 0$ on 1,204 zone-days in 14 zones, 609 of them in NO_2. Prices are not flat on these days. No intraday spread covered the battery's 15% round-trip loss. A capture share is undefined on such days, and they are excluded from the share outcomes along with days below each zone's 5th percentile of positive V^*.

Table 1: Descriptive statistics by bidding zone, sorted by mean forecast wind penetration.

Zone	Wind (%)	Solar (%)	V* (€)	TD mean	TD median	TD value-wtd	PS mean	FC mean	Novelty	V* = 0	Days
DK_1	58.4	8.8	162	77.0	86.0	83.7	64.8	78.6	0.247	29	2,814
DK_2	29.0	6.3	175	76.8	85.7	82.8	67.4	78.1	0.246	24	2,814
DE_LU	26.9	12.9	180	83.1	90.1	87.4	74.7	89.1	0.197	0	2,814
PT	25.5	7.0	107	74.7	87.2	84.1	66.9	78.8	0.198	103	2,820
ES	24.2	15.6	112	76.1	88.1	85.5	69.7	83.5	0.191	100	2,814
SE_4	22.1	3.6	169	75.3	84.8	81.6	65.2	75.3	0.260	50	2,814
GR	19.9	16.8	208	69.1	80.5	77.4	60.6	74.2	0.247	69	2,820
FI	16.1	1.1	148	67.5	78.7	73.3	53.7	66.5	0.298	33	2,820
NL	15.9	8.6	190	80.0	87.2	84.3	70.2	83.2	0.193	0	2,814
BE	15.0	9.0	175	80.6	86.9	83.1	70.3	82.5	0.193	0	2,814
AT	13.6	6.0	156	80.8	87.6	83.9	71.8	82.8	0.181	4	2,814
PL	11.5	8.7	158	71.2	82.5	81.2	59.2	75.0	0.188	39	2,814
NO_2	11.2	0.0	68	17.5	67.6	67.9	11.3	21.9	0.312	609	2,814
SE_3	10.7	1.6	130	72.4	83.6	81.6	61.9	72.1	0.261	85	2,814
EE	10.0	7.4	240	69.9	77.5	70.3	57.0	69.6	0.253	17	2,820
FR	9.7	5.3	151	81.4	87.5	83.6	72.6	83.8	0.188	0	2,814
CH	0.1	4.0	100	77.7	88.3	84.0	70.5	81.4	0.167	34	2,813
IT_NORD	0.1	5.9	106	75.9	85.7	79.8	69.3	82.3	0.177	8	2,814
CZ	–	5.4	174	80.3	88.4	83.9	71.7	83.3	0.181	0	2,814

Notes: Means over 2019-01-01 to 2026-09-20. Wind and solar: TSO day-ahead forecast output as a share of forecast load. V*: perfect-foresight arbitrage value of a 1 MW / 2 MWh battery at 85% round-trip efficiency, € per day. TD, PS, FC: capture shares (%) of the typical-day, persistence and forecaster operators, on days with V* $>$ 0 and above the zone's 5th percentile of positive V*; value-weighted TD is ΣV_TD / ΣV* over those days. The forecaster's sample starts in May 2019 for DE_LU. V* = 0 counts days on which no intraday spread covers the round-trip loss. CZ publishes no wind forecast and is outside the estimation sample.

The typical-day operator, which knows the recent calendar shape and nothing about the day, captures a mean of 73.4% of the ceiling on days where a share is defined. Weighted by value, it captures 81.0%. Mean shares are pulled down by low-value days on which small scheduling errors are large relative to the ceiling. NO_2 is the extreme case: its mean share is 17.5% against a median of 67.6%. Persistence captures 64.0%. An operator that trusts yesterday's prices does worse than one that averages the last four weeks, because a single day's idiosyncrasies are more often noise than signal. The forecaster captures 76.3% (83.0% value-weighted). On the days both operate, it earns €3.45 per MW per day more than the calendar operator, or 2.85 points of capture.

This is how much *this model* gains from the TSOs' fundamentals, not what the information is worth. The forecaster is trained on squared price error, which need not translate into arbitrage revenue.

Its gain is largest in Spain (+7.4 points), Italy North (+6.4) and Germany-Luxembourg (+6.0). In Finland, both Swedish zones and Estonia it does no better than the calendar (Figure 1).

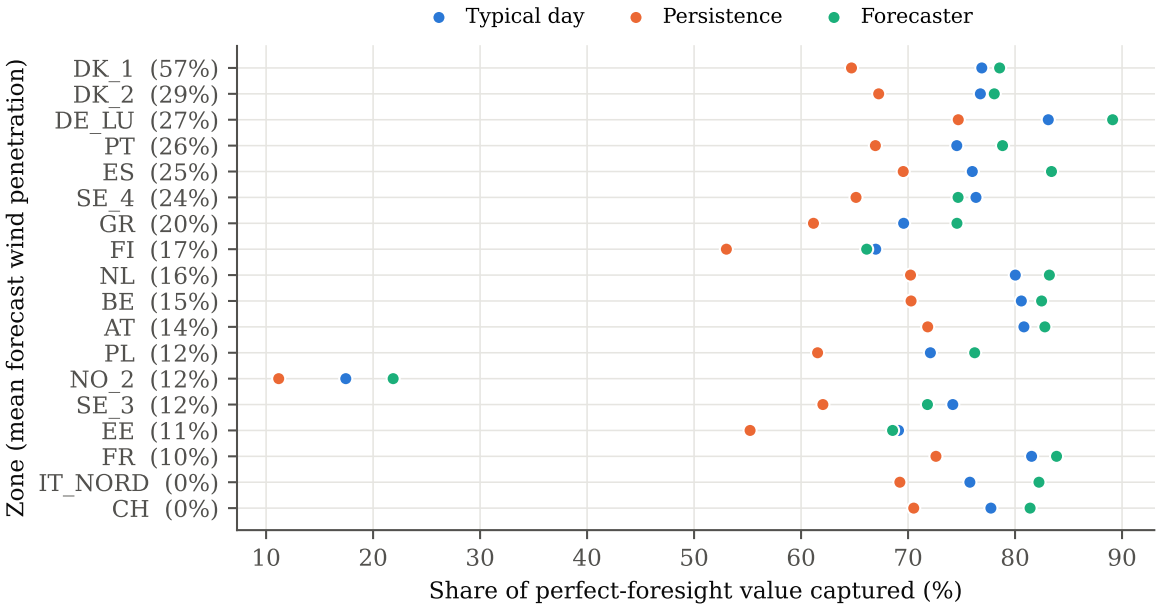


Figure 1: Share of the perfect-foresight value captured by each operator, by zone. Zones are ordered by mean forecast wind penetration, shown in brackets.

5.2 Which renewable creates arbitrage value

Table 2 reports the registered specification for every registered outcome.

Table 2: The effect of forecast wind and solar penetration on storage outcomes (registered).

	TD capture	PS capture	FC capture	log V*	Novelty	Spread (€)
Wind (per 10 pp)	-0.21	-0.89	-0.99	-0.020	0.013***	-2.96***
	(0.53)	(0.72)	(0.53)	(0.017)	(0.002)	(1.24)
WCB p-value	[0.744]	[0.372]	[0.184]	[0.256]	[<0.001]	[<0.001]
Solar (per 10 pp)	4.15	5.53	3.68	0.249***	-0.051***	22.14***
	(2.93)	(3.24)	(2.72)	(0.052)	(0.010)	(3.57)
WCB p-value	[0.186]	[0.116]	[0.205]	[<0.001]	[0.001]	[<0.001]
Wind − solar	-4.36	-6.42**	-4.67*	-0.269***	0.064***	-25.10***
	(2.59)	(2.71)	(2.40)	(0.057)	(0.011)	(4.19)
WCB p-value	[0.120]	[0.042]	[0.071]	[<0.001]	[<0.001]	[<0.001]
Mean of outcome	73.42	63.95	76.26	4.576	0.222	90.43
Zone-days	43,829	43,754	43,640	45,876	46,884	46,884
Zones	18	18	18	18	18	18

Notes: REGISTERED. Each column regresses the outcome on forecast wind and solar penetration with zone×year, zone×month and date fixed effects. Capture shares in percent of V*; novelty in correlation units; spread in €/MWh. log V* excludes the 1,008 zone-days with V* = 0 (see Table 6 for log(1+V*)). 'Wind − solar' is the wind coefficient minus the solar coefficient. Zone-clustered standard errors in parentheses; wild cluster restricted bootstrap p-values (Webb weights, 9,999 draws, fixed effects re-projected in every draw) in brackets, and stars use them. * p<0.1, ** p<0.05, *** p<0.01.

The registered scale outcome is log V^* (H3, registered without a predicted direction). Within a zone's year and season, and relative to other zones on the same day:

- **Solar creates arbitrage value.** 10 pp more forecast solar raises the day's perfect-foresight value by 28% (log V^* +0.249, p $<$ 0.001) and widens the daily spread by €22/MWh (p $<$ 0.001).
- **Wind does not.** Its effect on value is small and insignificant (log V^* −0.020, p = 0.26), and it *narrows* the spread by €3/MWh (p $<$ 0.001). The registered hypothesis text predicted that wind would "raise [value], or reshape it." The first half of that prediction failed.

The registered outcome drops the 1,008 zone-days in the estimation sample with $V^* = 0$. Keeping them changes nothing: log(1 + V^*) gives +0.245 for solar, and in euros solar adds €47.7 per MW per day (Table 13, exploratory). Figure 2 shows the relationship is close to linear across the range of residual penetration.

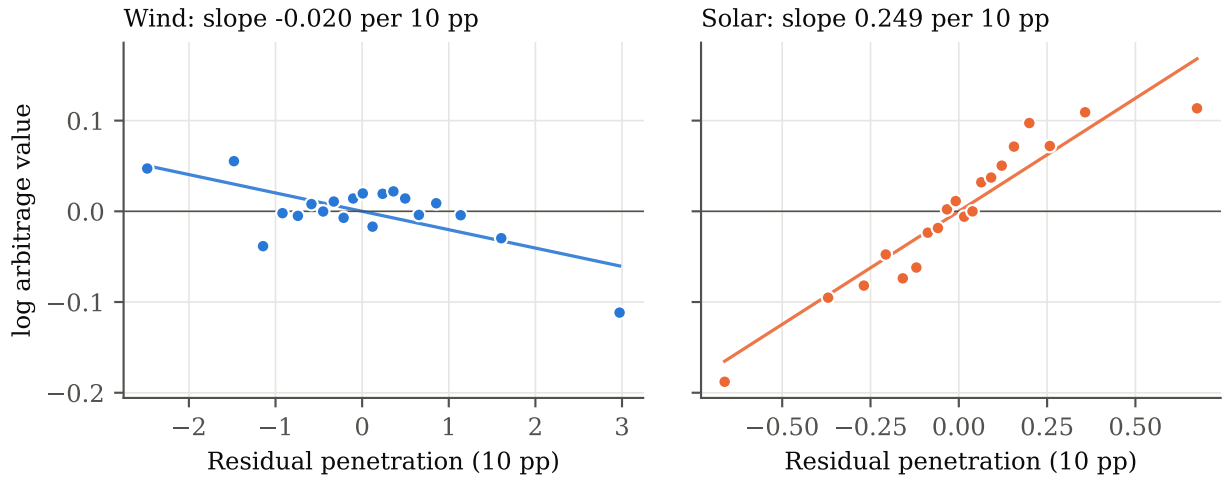


Figure 2: Binned scatter of log arbitrage value against residual forecast wind (left) and solar (right) penetration. Both variables are residualised on the other treatment and on zone×year, zone×month and date fixed effects, so the slope equals the regression coefficient in Table 2. Each point is a vigintile of the residualised treatment.

Figure 3 shows two days in Germany-Luxembourg in 2024, chosen mechanically before any regression as the days with the highest forecast wind and the highest forecast solar penetration. Neither day has an unusual *shape*. Each correlates closely with the typical day, with shape novelty of 0.086 and 0.088 against a zone-year mean of 0.193. What differs is the amplitude. On the windiest day prices sat low and flat, and the ceiling was €33 against a 2024 mean of €208. On the sunniest day

prices fell to −€60/MWh in the early afternoon, at the hours the typical day predicted, and the ceiling was €364. The figure illustrates the scale effect, not the shape effect. We return to shape next.

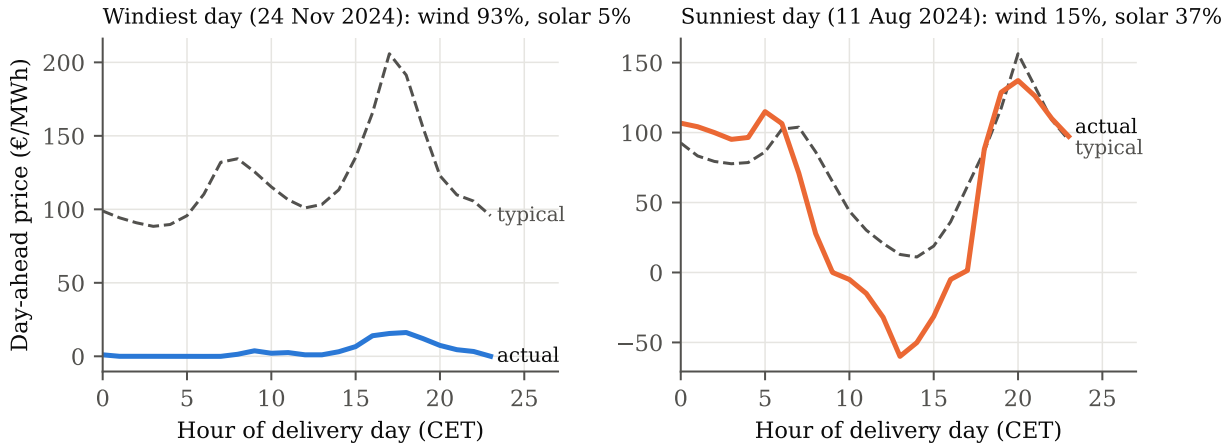


Figure 3: Two days in DE_LU, 2024: day-ahead prices (solid) against the typical-day profile of the previous 28 days (dashed). Left: the day with the highest forecast wind penetration. Right: the day with the highest forecast solar penetration.

5.3 The mechanism: how typical the day's shape is

Shape novelty, $1 - \text{corr}(\pi_d, \bar{\pi}_d^{TD})$, measures how far a day's price shape departs from the typical day. The registered hypothesis H2 predicted that wind raises it, and raises it more than solar does. Both parts hold (Table 2, column 5; Figure 4):

- **Wind makes the day less typical:** +0.013 per 10 pp, $p < 0.001$.
- **Solar makes it more typical:** −0.051 per 10 pp, $p = 0.001$.

Wind varies far more from day to day than solar. After the fixed effects, the residual standard deviation of wind penetration is 11.8 pp, against 2.9 pp for solar. Per standard deviation, the two effects are of similar size and opposite sign (+0.016 and −0.015, each about 7% of the mean novelty of 0.22). This comparison is exploratory.

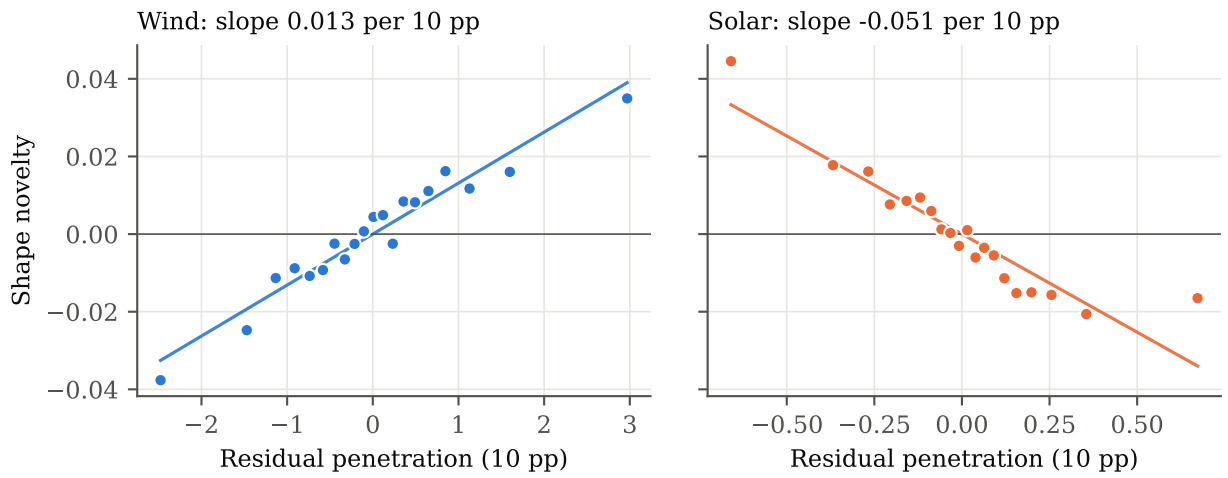


Figure 4: Binned scatter of shape novelty, 1 − corr(day's prices, typical-day profile), against residual forecast wind (left) and solar (right) penetration. Construction as in Figure 2.

A Pearson correlation with a fixed profile rises mechanically when the common diurnal component grows relative to noise. So part of "solar makes the shape more typical" could be "solar makes the shape bigger." Table 3 (exploratory) separates timing from amplitude:

- **A rank correlation**, which ignores amplitude, gives the same solar coefficient (−0.051) and a slightly smaller wind coefficient (+0.010).
- **Timing:** solar moves the day's price peak 0.53 hours and its trough 0.93 hours closer to the typical day's. Wind moves the peak 0.21 hours further away and has no effect on the trough.
- **Neighbours:** with region×date effects, which compare each zone only with its own coupling region, the solar coefficient falls to −0.037 and the wind coefficient to +0.012. Both remain significant.

Table 3: Shape: amplitude, rank and timing.

	Novelty (Pearson)	Novelty (rank)	Peak shift (h)	Trough shift (h)	Novelty, region×date
Wind (per 10 pp)	0.013***	0.010***	0.210***	-0.051	0.012***
	(0.002)	(0.002)	(0.025)	(0.044)	(0.002)
WCB p-value	[<0.001]	[<0.001]	[<0.001]	[0.248]	[0.002]
Solar (per 10 pp)	-0.051***	-0.051***	-0.530***	-0.934***	-0.037**
	(0.010)	(0.010)	(0.196)	(0.205)	(0.013)
WCB p-value	[0.001]	[<0.001]	[0.008]	[<0.001]	[0.017]
Wind − solar	0.064***	0.061***	0.741***	0.883***	0.049***
	(0.011)	(0.010)	(0.193)	(0.176)	(0.012)

	Novelty (Pearson)	Novelty (rank)	Peak shift (h)	Trough shift (h)	Novelty, region×date
WCB p-value	[<0.001]	[<0.001]	[<0.001]	[<0.001]	[0.002]
Mean of outcome	0.222	0.222	3.634	4.423	0.222
Zone-days	46,884	46,884	46,884	46,884	46,654
Zones	18	18	18	18	18

Notes: EXPLORATORY except the first column (REGISTERED, H2). Rank novelty uses Spearman rather than Pearson correlation with the typical-day profile, so it responds to timing but not amplitude; peak and trough shifts are the hour distances between the day's actual and typical maximum and minimum. The last column replaces date with region×date fixed effects. * p<0.1, ** p<0.05, *** p<0.01.

Solar therefore makes the day more calendar-like in timing, not just in amplitude. Two qualifications apply to the wind half, both from Section 6. The registered forecast-error placebo fails for novelty. And the next day's forecast wind predicts today's novelty about as strongly as today's does. Holding it fixed cuts the wind coefficient from 0.013 to 0.005 and leaves the solar coefficient unchanged. Solar's effect is a property of the day. Wind's effect on shape belongs largely to multi-day weather regimes.

5.4 Capture: the pre-registered test

Our key hypothesis (H1) was that wind lowers the share of value capturable from the calendar more than solar does, so $\beta_w < \beta_s$ for typical-day capture. **The test is inconclusive.** The difference has the predicted sign, -4.36 points of capture per 10 pp (standard error 2.59), with a WCB p-value of 0.120. The jackknife (CV3) p-value is 0.147. The exact Rademacher bootstrap, enumerating all 2^{18} sign patterns, gives 0.116.

The difference also decomposes against the hypothesis as registered, which said that wind lowers calendar capture. Wind does not move it (-0.21, $p = 0.74$). The whole difference comes from solar *raising* it ($+4.15$, $p = 0.19$). The same pattern, with more precision, holds for the other operators: -6.42 ($p = 0.042$) for persistence and -4.67 ($p = 0.071$) for the forecaster.

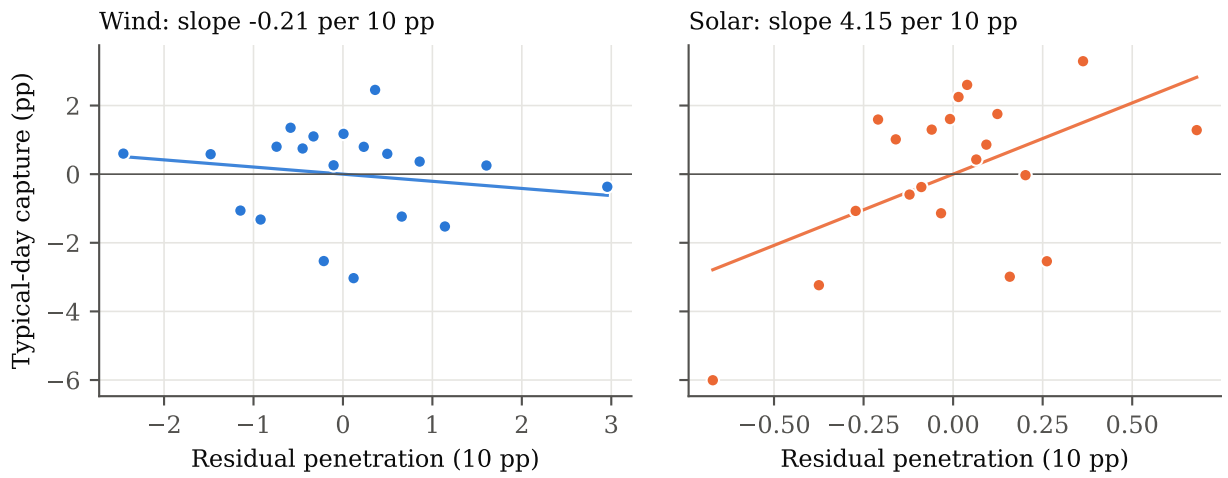


Figure 5: Binned scatter of typical-day capture share against residual forecast wind (left) and solar (right) penetration. Construction as in Figure 2.

The registered share outcome is badly conditioned, which we should have anticipated. A capture share divides by V^*, and the zone-relative floor admits days on which V^* is a few euros. In NO_2 the floor is €0.82 per day, and its capture shares range down to −6,613%. That one zone supplies most of the outcome's variance, and through the date effects it adds noise to the solar coefficient even though NO_2 has no solar. Dropping NO_2 moves the estimate to −6.45 (s.e. 1.44). We do not promote that or any other variant to the headline, since choosing among constructions after seeing results is exactly what registration exists to prevent. Section 6 reports them all.

Table 4 shows how the estimate moves as fixed effects are added (exploratory in presentation). With zone and zone×month effects only, the difference is −9.65 and conventionally significant at 1%. Adding zone×year effects leaves it at −8.75. Adding date effects halves it. Common shocks across Europe account for much of the unconditional association, which is why a single-market time series cannot answer this question. We have not tested which common shocks those are.

Table 4: How the estimate moves as fixed effects are added.

	None	Zone	+ Zone×month	+ Zone×year	+ Date (main)
Wind (per 10 pp)	-0.17	-0.59	-0.72**	-0.49	-0.21
	(0.51)	(0.42)	(0.31)	(0.39)	(0.53)
Solar (per 10 pp)	8.32**	7.04***	8.94***	8.25***	4.15
	(2.90)	(0.66)	(1.39)	(1.49)	(2.93)
Wind − solar	-8.49***	-7.63***	-9.65***	-8.75***	-4.36
	(2.55)	(0.54)	(1.16)	(1.30)	(2.59)
Mean of outcome	73.42	73.42	73.42	73.42	73.42
Zone-days	43,833	43,833	43,833	43,833	43,829
Zones	18	18	18	18	18

Notes: EXPLORATORY in presentation (the final column is the registered specification). Outcome: typical-day capture share (%). Fixed effects added left to right. Zone-clustered standard errors. Unlike Table 2, stars here use conventional cluster-robust p-values, which overstate significance with 18 clusters. * p<0.1, ** p<0.05, *** p<0.01.

5.5 The value that requires information

The capture share combines two things: the ceiling, and what the operator misses. Table 5 (exploratory) separates them in euros.

Table 5: The value that requires information, in euros (exploratory).

	V* − TD (€)	V* − FC (€)	FC − TD (€)	FC − TD (pp)
Wind (per 10 pp)	-0.50	0.12	-0.63*	-0.78***
	(0.55)	(0.61)	(0.19)	(0.18)
WCB p-value	[0.381]	[0.810]	[0.057]	[0.003]
Solar (per 10 pp)	-1.55	-2.47*	0.93	-0.48
	(1.24)	(1.20)	(0.61)	(0.61)
WCB p-value	[0.237]	[0.057]	[0.135]	[0.455]
Wind − solar	1.04	2.59*	-1.56**	-0.30
	(1.39)	(1.47)	(0.67)	(0.65)
WCB p-value	[0.464]	[0.098]	[0.026]	[0.661]
Mean of outcome	30.78	27.44	3.45	2.85
Zone-days	46,884	46,644	46,644	43,640
Zones	18	18	18	18

Notes: EXPLORATORY. Euro outcomes are per MW per day for the 2-hour battery: the value the typical-day and forecaster operators leave on the table, and the forecaster's gain over typical day. The last column is the difference in capture shares. Specification and inference as Table 2. * p<0.1, ** p<0.05, *** p<0.01.

- **The calendar operator's shortfall does not rise with either technology.** It leaves €30.8 per MW per day on the table on average. A windier day changes that by −€0.50 ($p = 0.38$), and a sunnier one by −€1.55 ($p = 0.24$). The two do not differ ($p = 0.46$). With weather instruments, the solar effect is larger and significant: the shortfall falls (Section 6.5).
- **Solar raises the calendar operator's revenue one-for-one with the ceiling.** Its revenue rises by €49.3 per 10 pp of solar ($p < 0.001$, Table 13), against €47.7 for the ceiling.
- **The forecaster's advantage falls on windy days.** Its gain over the calendar falls by 0.78 points of capture per 10 pp of wind ($p = 0.003$), or €0.63 per day ($p = 0.057$). Solar does not move it.

These results change the reading of H1. A share rises whenever the ceiling rises and the absolute shortfall does not, and that is what happens with solar. So the H1 difference is mostly a restatement of the scale result: solar adds value the calendar can find, and wind adds none. Neither technology creates value that only a forecaster can reach. The wind result is the opposite of what our motivation predicted. At the day-ahead stage the TSO wind forecast helps this forecaster *less* on

windy days. One interpretation is that windy days are when the day-ahead price already reflects the wind forecast most fully, leaving less for a model to add. We have not tested it.

Is this a property of our forecaster? The forecaster is trained on squared price error. Part of its effort therefore goes on the day's price level, which a battery that starts and ends each day empty barely trades on. If wind-driven value were informational but this model failed to extract it, a forecaster aimed at the day's shape should do better on windy days. We trained one (exploratory). It is the same model, trained on the within-day shape only, with each day's mean removed from the target. Table 6 reports it alongside a direct measure of what capture depends on: the rank correlation, across the day's hours, between each operator's plan and realised prices (Falezza 2026).

Table 6: A forecaster aimed at the day's price shape (exploratory).

	FC − TD (pp)	Shape − TD (pp)	Shape − TD (€)	Rank corr., TD	Rank gain, FC	Rank gain, shape
Wind (per 10 pp)	-0.78***	-0.22**	-0.18	-0.010***	0.002	0.001
	(0.18)	(0.09)	(0.15)	(0.002)	(0.001)	(0.001)
WCB p-value	[0.003]	[0.029]	[0.450]	[<0.001]	[0.345]	[0.599]
Solar (per 10 pp)	-0.48	-0.61	1.26*	0.051***	-0.013***	-0.012***
	(0.61)	(0.46)	(0.69)	(0.010)	(0.003)	(0.003)
WCB p-value	[0.455]	[0.185]	[0.091]	[<0.001]	[<0.001]	[<0.001]
Wind − solar	-0.30	0.39	-1.44*	-0.061***	0.015***	0.013***
	(0.65)	(0.46)	(0.68)	(0.010)	(0.003)	(0.003)
WCB p-value	[0.661]	[0.385]	[0.060]	[<0.001]	[<0.001]	[<0.001]
Mean of outcome	2.85	3.71	4.98	0.778	0.051	0.053
Zone-days	43,640	43,640	46,644	46,644	46,644	46,644
Zones	18	18	18	18	18	18

Notes: EXPLORATORY. 'Shape' is the forecaster trained on the within-day price shape (each day's mean removed from the target); FC is the registered forecaster. Capture gains are differences in capture shares (percentage points) over the typical-day operator; the euro gain is per MW per day. Rank correlation is Spearman's, across the day's hours, between an operator's plan prices and realised prices; rank gains are over the typical-day plan. Specification and inference as Table 2. * p<0.1, ** p<0.05, *** p<0.01.

- **The shape forecaster is the better operator.** On the same days it captures 77.3% of the ceiling (84.1% value-weighted), against 76.5% for the registered forecaster and 73.7% for the calendar. It earns €4.98 per MW per day more than the calendar, against €3.44 for the registered forecaster.
- **Its advantage does not rise on windy days either.** Its capture gain over the calendar falls by 0.22 points per 10 pp of wind ($p = 0.029$), and its euro gain does not move ($-€0.18$, $p = 0.45$).

- **The rank evidence shows why.** Wind lowers the calendar plan's rank correlation with realised prices (-0.010 per 10 pp, $p < 0.001$), and solar raises it ($+0.051$, $p < 0.001$). This is the shape result of Section 5.3 in the units capture depends on. But neither forecaster's rank advantage over the calendar rises with wind ($p = 0.35$ and 0.60). Windy days do scramble the order of the day's prices, and the day-ahead fundamentals do not recover that order.

So the absence of an informational premium to wind is not a quirk of one loss function. Two forecasters built on the TSOs' published fundamentals fail to find one.

5.6 Heterogeneity: wind's cost rests on one zone

The registered heterogeneity test (H4) interacts forecast wind penetration with K, the zone-year's mean wind penetration. The interaction is -0.35 points of typical-day capture per 10 pp of wind, per additional 10 pp of K (s.e. 0.08, WCB $p = 0.027$). H4 is confirmed as registered. Table 7 shows that it rests on a single zone.

Table 7: Heterogeneity in the wind effect by the system's wind share.

Outcome	Term	Estimate (s.e.)	WCB p	Zone-days
TD capture	Wind × K, all 18 zones (registered)	-0.35 (0.08)	0.027	43,829
TD capture	Wind × K, without DK_1	-0.41 (0.40)	0.626	41,203
TD capture	Wind × K, without DK_1 and DK_2	0.23 (0.47)	0.619	38,573
TD capture	Wind effect at DK_1's K (registered model)	-0.94 (0.34)	–	43,829
TD capture	Wind slope, lowest tercile of K	-1.47 (2.01)	–	43,829
TD capture	Wind slope, middle tercile of K	0.70 (0.47)	–	43,829
TD capture	Wind slope, highest tercile of K	-0.40 (0.56)	–	43,829
TD capture	Highest minus lowest tercile	1.07 (1.74)	0.810	43,829
FC − TD (pp)	Wind × K, all 18 zones	0.08 (0.07)	0.349	43,640
FC − TD (€)	Wind × K, all 18 zones	-0.13 (0.06)	0.113	46,644

Notes: Interaction of forecast wind penetration (per 10 pp) with K, the zone-year's mean wind penetration (per 10 pp, centred at the estimation-sample mean). The first row is REGISTERED (H4); all others are EXPLORATORY. Tercile cut-points are over zone-years: K below 11.8%, 11.8–22.5%, and above 22.5%. The FC − TD rows ask whether the forecaster's advantage over the calendar operator grows with a system's wind share. Zone-clustered standard errors; – where no bootstrap was run.

Denmark West (DK_1) is the only zone with K above 38%. Its zone-years run from 51% to 66%. It also carries a third of the identifying variation in wind (Table 12). At DK_1's mean K of 58%, the registered model implies that a windier day lowers calendar capture by 0.94 points per 10 pp (s.e. 0.34). At the estimation-sample mean ($K = 18.4\%$) the effect is $+0.46$ (s.e. 0.49). Without DK_1 the interaction is -0.41 with a standard error of 0.40 ($p = 0.63$). Without both Danish zones it changes sign. Splitting zone-years into terciles of K gives wind slopes of -1.47, $+0.70$ and -0.40 from the lowest tercile to the highest, with no monotone pattern (highest minus lowest $+1.07$, $p =$

0.81). The forecaster's advantage over the calendar does not grow with K either (the interaction is +0.08 points, p = 0.35, or −€0.13 per day, p = 0.11).

So the registered result is a statement about one system, not a pattern across systems. K is also not randomly assigned. High-wind systems differ in interconnection, hydro access and market design, any of which could modulate how wind weather reaches prices.

6 Robustness and limitations

6.1 Placebos

The registered forecast-error placebo. The day-ahead auction clears before the wind forecast error is realised, so the error should not affect any day-ahead outcome. Table 8 adds realised minus forecast wind to the main specification.

Table 8: Placebos: the realised wind forecast error, and the next day's forecasts.

Outcome	Forecast wind (main)	Wind forecast error	Equivalence p	Next day's wind	Next day's solar
TD	-0.21	0.05 (0.39)	0.341	-1.57 (0.31)	-1.31 (0.78)
capture		[0.909]		[0.002]	[0.082]
PS	-0.89	0.07 (0.49)	0.056	-1.70 (0.35)	-0.43 (0.77)
capture		[0.887]		[0.001]	[0.536]
log V*	-0.020	-0.003 (0.022)	0.221	-0.040 (0.007)	0.000 (0.021)
		[0.895]		[<0.001]	[0.985]
Novelty	0.013	0.009 (0.003)	0.098	0.016 (0.002)	0.004 (0.005)
		[0.015]		[<0.001]	[0.480]
Spread	-2.96	-2.53 (2.33)	0.427	-1.71 (0.59)	0.56 (2.00)
		[0.295]		[0.001]	[0.800]

Notes: The forecast-error column is REGISTERED; the lead columns are EXPLORATORY. Each placebo is added to the main specification. The forecast error is realised minus forecast wind; next day's wind and solar are day d+1's forecast penetration, published after day d's auction. Coefficient (zone-clustered s.e.) [WCB p-value]. Equivalence p is a two-one-sided test (EXPLORATORY) that the error coefficient lies within ± the absolute forecast-wind coefficient, with 17 degrees of freedom; a large value means the placebo cannot be distinguished from the effect it is meant to benchmark. Next-day forecasts exist at gate closure, so the lead columns test intertemporal dependence rather than serving as clean placebos; see the text.

The placebo fails for shape novelty. The error coefficient is 0.009 per 10 pp (p = 0.015), about two-thirds of the forecast coefficient. Elsewhere it does not reject, but that is weak evidence. The error coefficients are imprecise enough that the equivalence tests in Table 8 cannot show them to be smaller than the forecast effects they are meant to benchmark (equivalence p-values run from 0.06 to 0.43, so none establishes equivalence at 5%). For the spread, the error coefficient (−2.53) is as large as the forecast coefficient (−2.96).

Two readings of the novelty failure fit the data, and we cannot separate them. Market participants trade on their own weather forecasts, so the TSO's error may partly proxy for information the market had. Or large forecast errors may occur on frontal and transition days whose price shapes are atypical for reasons the market knew. Either way, the wind coefficient on novelty should not be read as the effect of the published forecast alone. With fixed effects and two correlated, mismeasured regressors, the direction of any bias in the other coefficients is not signed.

The next day's forecasts (exploratory). We added the next day's forecast penetration as a second placebo, on the premise that it cannot affect day d's auction. That premise is wrong, and the test is informative for that reason. Weather forecasts for $d+1$ exist when day d's auction clears. Reservoir hydro and multi-day unit commitment make prices respond to anticipated wind. Tomorrow's wind forecast predicts today's typical-day capture (-1.57, $p = 0.002$), today's value (-0.040, $p < 0.001$) and today's novelty ($+0.016$, $p < 0.001$). Tomorrow's solar forecast predicts almost nothing.

Table 9 uses the lead as a control instead. Within the fixed effects, today's and tomorrow's wind forecasts correlate at 0.50. Holding tomorrow fixed leaves every solar coefficient essentially unchanged: 0.243 for log V^* and -0.050 for novelty. It cuts the wind coefficient on novelty from 0.013 to 0.005 ($p = 0.19$). Solar acts on the day. Much of wind's effect on the day's shape belongs to multi-day weather regimes, which is also consistent with the novelty placebo's failure.

Table 9: Today's effects, holding tomorrow's forecasts fixed (exploratory).

Outcome	Wind	Wind, given tomorrow	Solar	Solar, given tomorrow	Zone-days
TD capture	-0.19	0.58 (0.66) [0.405]	4.20	4.63 (2.83) [0.118]	43,533
log V*	-0.020	-0.000 (0.017) [0.994]	0.250	0.243 (0.049) [<0.001]	45,564
Novelty	0.013	0.005 (0.002) [0.187]	-0.051	-0.050 (0.010) [<0.001]	46,566
Spread (€)	-2.93	-2.09 (1.15) [0.012]	22.24	21.66 (3.26) [<0.001]	46,566
V* − TD (€)	-0.50	-0.78 (0.63) [0.153]	-1.54	-1.94 (1.32) [0.186]	46,566

Notes: EXPLORATORY. Columns 2 and 4 repeat the main-specification coefficients on the same sample; columns 3 and 5 add the next day's forecast wind and solar penetration as controls. Coefficient (zone-clustered s.e.) [WCB p-value]. Within the fixed effects, today's and tomorrow's forecast wind penetration correlate at 0.50.

6.2 Robustness of the registered test

Table 10 reports the registered robustness set for H1 and, marked †, the exploratory checks the referee requested. We present them in full, without selecting.

Table 10: Robustness of the key estimate (typical-day capture).

Specification	β wind	β solar	β wind − β solar	p (diff)	Zone-days
Baseline (2h)	-0.21 (0.53)	4.15 (2.93)	-4.36 (2.59)	0.120	43,829

Specification	β wind	β solar	β wind − β solar	p (diff)	Zone-days
1h battery	-0.16 (0.49)	4.23 (2.58)	-4.40 (2.34)	0.090	43,826
4h battery	-0.22 (0.80)	4.96 (4.24)	-5.18 (3.65)	0.178	43,824
14-day typical day	-0.29 (0.58)	3.93 (3.08)	-4.22 (2.68)	0.141	43,829
Ratio floor 1st pct	-0.34 (0.99)	4.07 (4.60)	-4.41 (3.89)	0.296	45,464
Ratio floor 10th pct	-0.06 (0.33)	4.66 (2.19)	-4.73 (1.97)	0.036	41,700
Excluding gas crisis	-0.06 (0.33)	4.91 (1.78)	-4.97 (1.64)	0.009	31,434
Two-way cluster (zone, date)	-0.21 (0.52)	4.15 (2.85)	-4.36 (2.52)	0.102	43,829
Region × date FE †	-0.59 (0.74)	2.94 (3.41)	-3.54 (2.97)	0.294	43,311
Zone × year × month FE †	-0.01 (0.51)	6.88 (2.07)	-6.89 (1.78)	0.003	43,829
Fixed load denominator †	-0.20 (0.50)	6.40 (2.84)	-6.60 (2.53)	0.027	43,829
Before 15-min MTU †	-0.20 (0.58)	3.59 (3.11)	-3.79 (2.79)	0.205	37,607
Excluding negative-price days †	-0.40 (0.59)	4.90 (3.19)	-5.30 (2.77)	0.087	39,498

Notes: Outcome: typical-day capture share (%), per 10 pp of forecast penetration. Rows without a dagger are the REGISTERED robustness set; rows marked † are EXPLORATORY. Unless stated, zone×year, zone×month and date fixed effects. p-values are WCB (Webb, 9,999 draws) except the two-way-clustered row (conventional).

- **Point estimates** of $\beta_w - \beta_s$ range from −3.54 (region×date effects) to −6.89 (zone×year×month effects). All have the predicted sign.
- **p-values** range from 0.003 to 0.296. Among the registered checks, three are below 0.10 (the 1-hour battery, the 10th-percentile floor and excluding the gas crisis), and four are not (the 4-hour battery, the 14-day window, the 1st-percentile floor and two-way clustering).
- **Leave-one-zone-out estimates** range from −3.08 (without Poland) to −6.45 (without NO_2). This is weak evidence of stability, since every pair of estimates shares 16 of 18 clusters.

Some variants are more precise for reasons unrelated to the sign. Excluding the gas crisis removes 28% of zone-days yet cuts the standard error by 37%, which shows that crisis days are high-leverage in the ratio. Nothing in Table 10 changes the registered verdict: the sign is as predicted, and the test is inconclusive.

Where in the distribution. The capture share is capped at 100 and unbounded below, so the mean leans on bad days. Table 11 (exploratory) estimates the quartiles with the two-step estimator of Canay (2011), which removes the fixed effects as location shifts before running the quantile regression. The wind-minus-solar difference is −4.66 at the lower quartile, −2.10 at the median and −0.59 at the upper quartile, and none is significant. Wind is close to zero at every quartile. Whatever difference there is lives in the lower tail: solar mainly reduces the calendar operator's bad days, rather than shifting its typical day. One caveat applies. The estimator's location-shift assumption is strongest when each fixed effect is estimated from many observations, and each date effect is estimated from at most 18.

Table 11: Typical-day capture at its quartiles (exploratory).

Estimator	β wind	β solar	β wind − β solar
Mean (OLS, Table 2)	-0.21 (0.53)	4.15 (2.93)	-4.36 (2.59)
Quantile 0.25	-0.15 (0.43)	4.51 (2.94)	-4.66 (2.68)
Quantile 0.50	-0.04 (0.51)	2.06 (3.31)	-2.10 (3.01)
Quantile 0.75	-0.16 (0.70)	0.43 (3.93)	-0.59 (3.63)

Notes: EXPLORATORY. Outcome: typical-day capture share (%), per 10 pp. Quantile rows use the two-step estimator of Canay (2011): the fixed effects are estimated by OLS and removed as location shifts, then the quantile regression is run. Standard errors from a pairs cluster bootstrap over zones (199 draws, both steps re-estimated). The mean row uses zone-clustered standard errors.

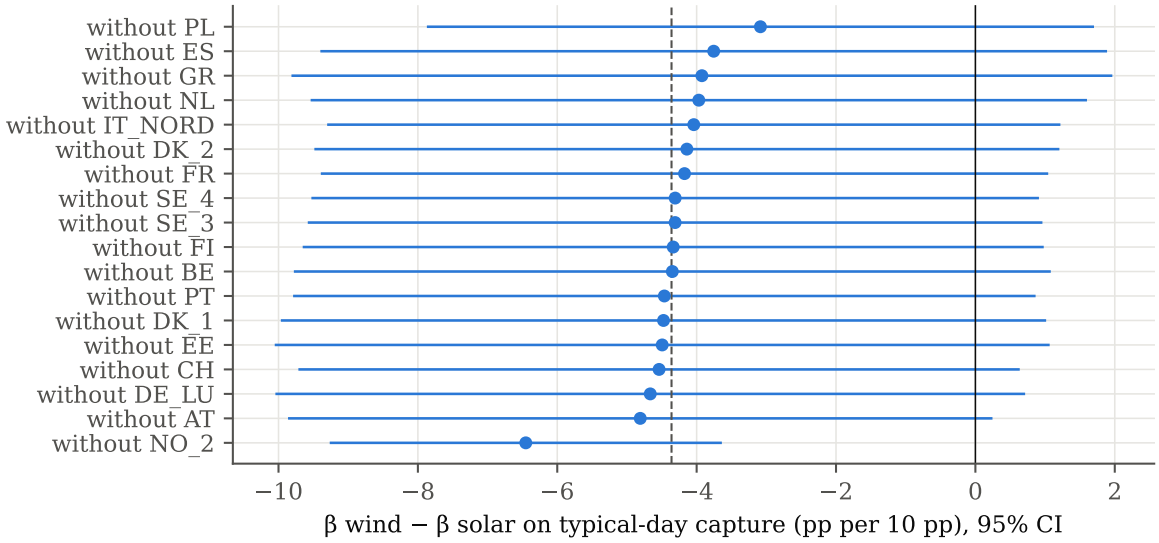


Figure 6: Leave-one-zone-out estimates of $\beta_w - \beta_s$ for typical-day capture, with 95% confidence intervals from zone-clustered standard errors. The dashed line is the full-sample estimate.

Inference. Eighteen clusters of unequal influence are the setting in which cluster-robust inference is least reliable (MacKinnon and Webb 2017; MacKinnon, Nielsen, and Webb 2023). Table 12 reports each zone's partial leverage. For wind, Denmark West contributes a third of the identifying variation. For solar, the largest share is Greece's, at 15%. NO_2 has no solar at all, yet it contributes 2.4% of the solar variation, because date effects turn its zeros into deviations from other zones' sunny days. We therefore report three p-values for H1, and they agree: 0.120 (Webb WCB), 0.147 (CV3 jackknife) and 0.116 (exact Rademacher enumeration). Two-way clustering by zone and date gives a slightly *smaller* standard error (2.52 against 2.59). Driscoll–Kraay standard errors, which allow any dependence across zones on the same day, are much smaller still (0.86 with a 7-day lag and 0.97 with a 28-day lag, against 2.59). They allow serial dependence only within a lag window, however, and zone-level persistence beyond it is exactly what zone clustering protects against. We do not rely on them.

Table 12: Partial leverage of each zone (exploratory).

Zone	Partial leverage, wind	Partial leverage, solar	Zone-days
GR	0.075	0.151	2,553
DK_1	0.328	0.122	2,626
ES	0.060	0.092	2,556
DE_LU	0.058	0.086	2,668
NL	0.031	0.082	2,672
BE	0.032	0.075	2,673
EE	0.025	0.071	2,154
PL	0.016	0.053	2,209
AT	0.046	0.048	2,669
CH	0.018	0.041	2,638
PT	0.090	0.033	2,538
DK_2	0.075	0.033	2,629
IT_NORD	0.018	0.033	2,610
NO_2	0.016	0.024	2,091
FI	0.046	0.019	2,474
FR	0.018	0.017	2,647
SE_4	0.038	0.010	1,706
SE_3	0.011	0.010	1,720

Notes: EXPLORATORY. Share of the residualised variation in each treatment (within the fixed effects, partialled on the other treatment) that each zone contributes, in the typical-day capture sample. Equal clusters would each have 0.056. Sorted by solar.

6.3 Robustness of the scale result

Table 13 (exploratory) subjects "solar raises V^*" to the same checks. The coefficient ranges from 0.148 (region×date effects) to 0.293 (a fixed load denominator), and every p-value is below 0.01. It survives excluding the roughly 10% of zone-days with a negative price hour, which rules out the concern that solar raises V^* only by producing negative prices a battery is paid to absorb. Wind is not significant at 5% in any row.

Table 13: Robustness of the effect on arbitrage value (exploratory except the first row).

Outcome / specification	β wind [WCB p]	β solar [WCB p]	Zone-days
log V* (registered)	-0.020 [0.256]	0.249 [<0.001]	45,876
log(1 + V*)	-0.017 [0.398]	0.245 [<0.001]	46,884
V* in €	-3.21 [0.088]	47.73 [<0.001]	46,884
V_TD in €	-2.70 [0.105]	49.27 [<0.001]	46,884
log V*, region × date FE	-0.014 [0.560]	0.148 [0.009]	45,577

Outcome / specification	β wind [WCB p]	β solar [WCB p]	Zone-days
log V*, zone × year × month FE	-0.020 [0.268]	0.288 [<0.001]	45,876
log V*, fixed load denominator	-0.019 [0.304]	0.293 [<0.001]	45,876
log V*, no negative-price days	-0.012 [0.522]	0.263 [<0.001]	41,493

Notes: The first row is REGISTERED (H3, no directional prediction); the rest are EXPLORATORY. Per 10 pp of forecast penetration. log(1+V*) and V* in € keep the zero-value days that log V* drops. V_TD is the typical-day operator's revenue; it is reported in euros because it is zero or negative on 7% of zone-days, which a log would drop.

6.4 Threats to identification

The fixed effects are additive. Zone×month effects are held fixed from 2019 to 2026, so they cannot absorb a solar seasonal amplitude that grows with installed capacity. Replacing them with zone×year×month effects moves every solar coefficient *away* from zero (Tables 10 and 13). The additive specification is conservative here, but it is also fragile to this natural refinement.

Market coupling. Date effects remove only what is common to all 18 zones. Part of the identifying variation therefore compares different regions on the same day, for example sunny Iberia against a cloudy, hydro-priced Nordic region. Those regions differ in marginal technology, interconnection and congestion as well as weather. Region×date effects compare each zone only with its own coupling region. They cut the solar coefficient on log V^* by about 40%, to 0.148 ($p = 0.009$), and H1 to -3.54 ($p = 0.29$). Coupling also violates the stable-unit assumption, because a zone's price responds to its neighbours' weather (Lago et al. 2018). Adding the neighbours' mean forecast penetration as a control (the solar coefficient on log V^* moves from 0.252 to 0.231 on the common sample, and the wind coefficient on novelty from 0.013 to 0.012; Table 14) does not change the conclusions. The coefficients are best read as the effect of a zone's own weather given its neighbours', estimated partly from comparisons across regions.

Table 14: Load and spillover controls (exploratory).

Specification	TD capture: wind	TD capture: solar	log V*: wind	log V*: solar	Novelty: wind	Novelty: solar
Baseline	-0.23	4.16 [0.188]	-0.021	0.252	0.013	-0.050
	[0.719]		[0.246]	[<0.001]	[<0.001]	[0.001]
+ log load	-0.17	5.05 [0.119]	-0.020	0.268	0.013	-0.055
forecast	[0.809]		[0.277]	[<0.001]	[<0.001]	[<0.001]
+ neighbours'	-0.35	4.07 [0.203]	-0.017	0.231	0.012	-0.048
penetration	[0.600]		[0.312]	[<0.001]	[<0.001]	[0.001]

Notes: EXPLORATORY. Coefficient per 10 pp [WCB p-value], main fixed effects, on the common sample where the load forecast and neighbours' penetration are defined (43,635 zone-days for TD capture). Neighbours' penetration is the mean forecast penetration of the other zones in the same coupling region on the same day.

The denominator moves with weather. Penetration is forecast output over forecast load, and forecast load itself falls on sunny days (-0.045 log points per 10 pp of solar, s.e. 0.007). Temperature is one reason. Embedded solar is another: the regulation defines total load to include demand met by distribution-connected generation (ENTSO-E 2020), but practice differs. The German regulator describes its consumption series as covering only electricity drawn from the grid, which would leave out rooftop solar consumed on site (Bundesnetzagentur, n.d.). Energinet receives no generation data from small Danish rooftop systems and estimates national solar output from a sample of metered plants (Energinet, n.d.). Elia estimates unmeasured decentralised generation and adds it back (Elia, n.d.). For the other zones we found no public statement either way. A fixed denominator (the zone-year's mean load) and a log load control both leave the results intact or stronger (Tables 10, 13 and 14), and the weather instruments below do not depend on the load forecast at all.

Forecast timing. The wind and solar series we use is the platform's day-ahead vintage, which Regulation 543/2013 requires by 18:00 on D$-$1. That is six hours after the auction, so the published forecast may contain information the auction did not have. The load forecast is due two hours before gate closure, but TSOs must update it after any significant change, and the platform serves only the latest revision. The platform keeps no record of when a forecast was submitted or revised. We confirmed this in its documentation (ENTSO-E 2020) and in the raw responses, whose only timestamp is the time of the query. So publication times cannot be recovered from this source. We found no evidence that any TSO nets expected curtailment at negative prices out of its renewable forecast, but we cannot rule it out. Excluding negative-price days leaves the results unchanged (Tables 10 and 13). The weather instruments address the timing question directly, because realised weather cannot respond to prices.

6.5 Identification from weather

The load denominator, the forecast's timing and possible curtailment netting are all channels through which the treatment could respond to something other than weather. Realised weather has none of them. We therefore instrument the two forecast penetrations with ERA5 reanalysis weather (exploratory). The data come from Open-Meteo's archive at 105 fixed points, five to eight per zone, chosen to sit in each zone's main wind and solar areas, with offshore clusters where a zone has them. Each point's role (wind, solar or both) and the reason for its location are recorded in the database.

The wind instrument is the capacity factor of a notional turbine at 100 m: cut-in 3 m/s, rated 12 m/s, cut-out 25 m/s. The solar instrument is the day's shortwave radiation. Each is averaged over the zone's points and the delivery day's hours, and expressed relative to its zone-year mean times the zone-year's mean forecast penetration. That puts it in treatment units, and its only variation within a zone-year comes from weather. Both first stages are strong: a coefficient of 0.67 (s.e. 0.06) of wind penetration on wind weather, with a joint F-statistic of 80, and of 0.97 (s.e. 0.02) of solar penetration on solar weather, with F = 1,030.

Table 15: Forecast penetration instrumented by realised weather (exploratory).

Outcome	β wind (IV)	β solar (IV)	β wind − β solar (IV)	Wind weather: RF p	Solar weather: RF p	Zone-days
TD capture	-0.13 (0.56)	6.51 (3.87)	-6.64 (3.49)	0.767	0.126	43,833
log V*	-0.030 (0.019)	0.287 (0.054)	-0.317 (0.062)	0.120	<0.001	45,876
Novelty	0.013 (0.003)	-0.069 (0.012)	0.082 (0.011)	0.006	<0.001	46,884
V* − TD (€)	-0.99 (0.71)	-5.00 (2.23)	4.01 (2.07)	0.128	0.032	46,884
FC − TD (pp)	-0.74 (0.17)	-1.03 (0.91)	0.28 (0.89)	0.005	0.287	43,644

Notes: EXPLORATORY. Forecast wind and solar penetration instrumented by ERA5 reanalysis weather at fixed points in each zone: a notional turbine's capacity factor at 100 m and daily shortwave radiation, each scaled by its zone-year mean penetration. Main fixed effects; zone-clustered standard errors in parentheses. RF p is the WCB p-value of the weather instrument in the reduced form (the outcome regressed on the instruments), which is robust to weak instruments. First stage: wind penetration on wind weather 0.67 (s.e. 0.06), joint F = 80; solar penetration on solar weather 0.97 (s.e. 0.02), joint F = 1030.

Table 15 reports 2SLS estimates, together with weak-instrument-robust reduced-form tests: the outcome regressed on the weather itself, with WCB p-values.

- **Solar raises value; wind does not.** Instrumented, 10 pp more solar raises log V^* by 0.287 (s.e. 0.054), against 0.249 with forecasts alone, and the reduced-form p-value is below 0.001. Wind's is 0.12.
- **The shape effects survive.** Wind raises novelty by 0.013 and solar lowers it by 0.069 per 10 pp, with reduced-form p-values of 0.006 and below 0.001.
- **The value that requires information still does not rise with wind.** The forecaster's gain over the calendar falls with wind, by 0.74 points per 10 pp (reduced-form p = 0.005), as in Table 5. One result is sharper than with forecasts. The calendar operator's euro shortfall *falls* with solar, by €5.00 per 10 pp (reduced-form p = 0.032). Solar weather makes the calendar operator miss less in euros, not just a smaller share.
- **The registered test remains inconclusive.** The instrumented wind-minus-solar difference in calendar capture is −6.64 (s.e. 3.49), larger than the forecast-based −4.36. The solar reduced-form p-value is 0.13.

The weather instruments therefore support every conclusion in Section 5 from a source that cannot respond to prices. They do not resolve the market-coupling question, which concerns which zones are compared rather than how the treatment is measured.

6.6 Limitations

Day-ahead only. Batteries earn much of their revenue in intraday, balancing and ancillary-service markets, which we do not model. Intraday markets are where forecast errors are traded, so any informational premium to wind is more likely to appear there.

Price-taking and daily independence. The battery does not move prices, which suits one battery and not a fleet (Butters, Dorsey, and Gowrisankaran 2025; Karaduman 2023). Each day starts and ends empty, which forgoes multi-day arbitrage. That understates value most for the four-hour battery and in hydro-dominated zones.

Two forecasters, not the frontier. Both forecasters are reasonable gradient-boosted models built on the TSOs' published fundamentals (Lago et al. 2021). Neither is decision-focused in the strict sense of being trained on arbitrage revenue itself. A commercial operator with its own weather models, or a model trained end to end on revenue, might find an informational premium to wind that ours do not. Our value-of-information results bound what these models extract, not what all information is worth.

Heterogeneity is descriptive. Comparisons across systems by wind share confound wind capacity with everything else that differs between those systems, and our one clear case is a single zone.

7 Conclusion

We set out to test whether wind and solar create storage value of different kinds: value that any operator who knows the calendar can capture, and value that only an operator who forecasts the day can reach. The data support a simpler and, for investors and regulators, more useful conclusion. The renewable mix changes how much day-ahead storage value there is. We find no evidence that it changes who can capture it.

The evidence has three layers, and they differ in strength.

- **Strongest: solar creates value, wind does not.** This was registered as an outcome without a predicted direction. It is large, precise, and survives every check in Section 6, including instruments built from realised weather. The check that most weakens it compares zones only with their neighbours, and it still leaves about 60% of the effect. It fits what is known about solar's effect on the intraday profile (Bushnell and Novan 2021) and extends it from prices to storage revenue.
- **Solid, with a qualification: solar makes the day more calendar-like.** It moves the daily peak and trough towards the hours the calendar predicts, by rank as well as by correlation. Wind's opposite effect on shape is registered and significant, but it is largely a multi-day weather effect, and its placebo test fails.
- **Negative: the value that requires information does not respond.** The calendar operator's euro shortfall does not rise on windy days, and with weather instruments it falls on sunny ones. Neither of two forecasters gains more on windy days. The registered capture-share

test (H1) has the predicted sign but is inconclusive, and in euros it reduces to the scale result. The registered heterogeneity result (H4) is Denmark West.

This is not the paper we set out to write. The earlier single-market analysis that motivated it found that decarbonisation shifts storage value towards sophisticated operators, and that claim did not survive year fixed effects. The first draft of this paper made a weaker version of the same claim, and it did not survive review. What survives is less dramatic and better supported. For day-ahead arbitrage, the case for storage in a solar-heavy system rests on value that simple schedules can capture. We found no premium to forecasting sophistication that grows with wind.

Implications, stated with their limits. These results concern the day-ahead market and a price-taking battery.

- An investor modelling day-ahead arbitrage revenue in a system adding solar can treat most of the increment as capturable by simple schedules. The share it will realise is better predicted by the level of value than by the generation mix.
- We find no evidence that wind-heavy systems favour operators with scale in forecasting at the day-ahead stage. That does not rule it out in the intraday and balancing markets where forecast errors are traded, which we do not model.

Extensions. Three follow from this design:

- **Intraday and balancing markets,** where wind's forecast errors are traded, and where any informational premium to wind is most likely to appear.
- **Weather forecasts as of gate closure.** Our instruments use realised weather. Archived numerical-weather-prediction forecasts issued before the auction would measure exactly what the market could have known, and separate the forecastable part of weather from the forecast error.
- **An equilibrium treatment of storage entry.** A fleet large enough to flatten the solar trough would change the shape it trades on (Butters, Dorsey, and Gowrisankaran 2025; Karaduman 2023).

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A Pre-registration, deviations and revision record

A.1 Registration

All hypotheses, the specification, the sample, the outcome definitions and the robustness set were fixed in a pre-registration document committed to the project's version control before any panel backtest had run (commit `6aa7cdc`, 27 September 2026).

	Outcome	Registered prediction	Result
H1	Typical-day capture	$\beta_w < \beta_s$	Predicted sign; not significant (WCB p = 0.120)
H2	Shape novelty	$\beta_w > 0$ and $\beta_w > \beta_s$	Holds ($p < 0.001$), with the placebo caveat in Section 6
H3	log V^*	No direction registered	Solar raises V^*; wind does not
H4	Typical-day capture	Wind effect more negative where wind is a larger share	Holds in the full sample (WCB p = 0.027), but driven by DK_1 alone
Placebo	All of the above	Forecast-error coefficient zero	Fails for novelty; elsewhere too imprecise to be informative

The registered hypothesis text also predicted that wind would "raise [arbitrage value], or reshape it." Wind does not raise value (Table 2), so that part of the hypothesis failed.

A.2 Deviations

Five deviations were logged before the first regression was estimated. Deviations 4 and 5 were logged after the ceiling V^*, which is an outcome, had been computed and inspected in summary statistics. The first draft said all deviations predated every outcome, which was not accurate for these two.

1. **Czechia leaves the regression sample.** It publishes no day-ahead wind forecast.
2. **The gas-crisis window is fixed at 1 July 2021 to 30 June 2023.** The registration named the check but not its dates.
3. **The forecaster predicts deviations from the typical-day profile, not price levels.**
4. **Capture shares require $V^* > 0$,** with the 5th-percentile floor computed among positive days. As logged, this deviation wrongly stated that only NO_2 has zero-value days and that its

prices are flat on them. In fact 1,204 zone-days across 14 zones have $V^* = 0$, and none has flat prices. The rule is unchanged; its description is corrected, and it changes the floor in 14 zones, not one.

5. **Implausible load forecasts are set to missing.** These are days outside 50–200% of the zone's median for the month. The rule affects 11 days.

A.3 Revision after review

A referee report on the first draft identified the following. Each was addressed as described.

- **A bootstrap implementation error.** Each draw's outcome was not re-projected off the fixed effects. Date effects cross the zone clusters, so this understated every wild-bootstrap p-value. H1's p-value moves from 0.099 to 0.120. The corrected code reproduces the referee's replication exactly.
- **Overclaiming.** The first draft's title, abstract and conclusion presented H1 as supporting a claim that solar-created value is "public" and wind-created value "informational." The value-of-information outcomes added in revision (Table 5) contradict the wind half of that claim, so it has been removed.
- **A missing registered check.** The 14-day typical-day window is now reported (Table 10).
- **Factual errors** about zero-value days, the fixed-effect structure (additive, not zone×year×month), the CET offsets, and the discussion of the example days (now Figure 3) have been corrected.
- **Identification checks added at the referee's request,** all exploratory:
  - region×date and zone×year×month fixed effects;
  - a fixed load denominator, a log load control, and neighbours' forecast penetration;
  - excluding negative-price days;
  - a pre-15-minute-MTU sample;
  - a lead placebo, which turned out to test intertemporal dependence instead;
  - equivalence tests for the forecast-error placebo;
  - rank-based novelty and timing measures;
  - H4 without the Danish zones, by terciles of K, and on the forecaster's gain;
  - value-weighted and median capture, and the calendar operator's revenue in euros;
  - jackknife standard errors, exact Rademacher enumeration, partial leverage by zone, and Driscoll–Kraay standard errors.
- **Requests first deferred, then carried out.** The first revision listed five requests as limitations. All five have since been addressed, each as an exploratory analysis:
  - a forecaster aimed at the day's price shape, with rank correlations (Table 6);
  - median and quartile regressions (Table 11);
  - ERA5 weather instruments for the treatments (Table 15);
  - documentary evidence on load netting and forecast timing (Section 6.4). Forecast publication times cannot be recovered: the platform does not record them.
- **Code changed alongside the first results.** The referee asked what changed in the analysis code after the first estimates were seen. In the commit that recorded the first results

(`8e1dc7f`), every change to `results.py` was figure code: layout, label placement, and the addition of Figure 3. No estimation code changed. Every later change is listed in this appendix and in the pre-registration's correction log.

- **Two forecaster feature bugs.** A rolling window counted rows rather than calendar days, and daily shares summed partial-coverage days. Neither leaked future information. Both were fixed and the forecaster re-run.
- **Literature** was added on intraday price shape, weather-based identification and cluster-robust inference. Each reference was verified before being cited.

Why the design is built this way. An earlier single-market analysis of this question on British data reported that decarbonisation shifts storage value towards sophisticated operators. That claim did not survive the addition of year fixed effects. The current design responds in three ways:

- identification comes from day-to-day variation within zone-years;
- the treatments are forecasts;
- the specification was fixed in advance.

The review of this paper found an error of a different kind, in the inference rather than the specification. That is why the code, data and review are all part of the record.

B Operator details

The optimisation. All operators solve the same mixed-integer programme with the HiGHS solver through SciPy, on the price vector their information set provides. The schedule is then settled at actual prices. The panel required approximately 700,000 solves with no solver failures. Each schedule was checked against the physical limits on state of charge and power before being recorded.

The forecaster. For each zone, a histogram gradient-boosted regression tree predicts, for every hour of day d, the price's deviation from the typical-day profile. Its features are:

- the market hour, weekday and month;
- the typical-day price for that hour;
- yesterday's price at that hour and yesterday's mean, each as a deviation from the profile;
- the TSO's day-ahead forecasts of load, wind and solar for that hour, each also expressed relative to its mean over the previous 28 calendar days;
- the day's forecast wind and solar shares of load, defined only when the forecasts cover every hour.

The model is refitted monthly on an expanding window, and it forecasts only months strictly after its training data. Germany-Luxembourg's zone begins only in October 2018, and the first model needs 180 days of training, so its forecasts begin on 1 May 2019.

The shape forecaster. It is identical to the forecaster above except for its target. That target is the hourly deviation from the typical-day profile, with each day's own mean removed, so that the model is trained only on the day's shape. The plan price is still the typical-day price plus the

model's prediction. Re-running the registered forecaster with the same code reproduced every stored daily revenue exactly.

Solver tolerance. On fewer than 20 of about 160,000 zone-day-battery cases per operator, an operator's settled revenue exceeds the perfect-foresight value by at most €0.015 per MW per day. This is within the MILP solver's default optimality tolerance, and it affects no reported number at the precision shown.

Reproducibility. Every table, figure and quoted number is produced by code in the project repository from the raw ENTSO-E data, and the paper is compiled from those outputs:

```
python -m gbmo.ingest.entsoe && python -m gbmo.ingest.load_zones
python -m gbmo.arbitrage.panel && python -m gbmo.arbitrage.panel --td14
python -m gbmo.arbitrage.panel_forecast && python -m gbmo.arbitrage.panel_forecast --target sha
python -m gbmo.ingest.weather_panel && python -m gbmo.analysis.weather_days
python -m gbmo.analysis.panel_data && python -m gbmo.analysis.results
python paper/build.py
```